

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I **P Dinesh Reddy / 192525399** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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Analytical Problem Solving

1. A cloud datacenter host has 64 CPU cores and supports CPU overcommitment at a ratio of 2.5:1. Each virtual machine (VM) requires 4 vCPUs. Calculate the maximum number of VMs that can be deployed on this host. If actual CPU utilization reaches 85%, analyze whether further VM allocation is advisable.

Solution

Step 1: Calculate total virtual CPUs available

- Physical CPU cores = **64**
- Overcommitment ratio = **2.5 : 1**

Total available vCPUs

$$= 64 \times 2.5$$

= **160 vCPUs**

Step 2: Calculate maximum number of VMs

Each VM requires **4 vCPUs**

Maximum VMs

= $160 \div 4$

= **40 VMs**

Answer: Maximum **40 VMs** can be deployed.

Step 3: Analysis

Actual CPU utilization = **85%**

An 85% utilization indicates that the host is already heavily loaded.

Further VM allocation may result in:

- Increased CPU contention
- Higher response time
- Performance degradation
- Reduced Quality of Service (QoS)
- Possibility of SLA violations

Conclusion

Although the host supports **40 VMs**, allocating additional VMs when CPU utilization is already **85% is not advisable**. The administrator should monitor resource usage or migrate workloads to another host.

2. A virtualized storage system provides 20 TB of physical storage with thin provisioning at a ratio of 3:1. Each VM is allocated 500 GB of virtual disk space. Calculate the maximum number of VMs that can be provisioned.

Solution

Step 1: Calculate virtual storage

Physical storage = **20 TB**

Thin provisioning ratio = **3 : 1**

Virtual storage

= 20×3

= **60 TB**

Step 2: Convert TB to GB

60 TB

$= 60 \times 1000 \text{ GB}$

$= 60000 \text{ GB}$

Step 3: Calculate number of VMs

Each VM requires **500 GB**

Maximum VMs

$= 60000 \div 500$

$= 120 \text{ VMs}$

Answer

The storage system can provision **120 virtual machines**.

Analysis

Thin provisioning allows allocating more virtual storage than physically available because not all VMs use their entire allocated space simultaneously.

Advantages:

- Better storage utilization
- Reduced storage cost
- Efficient resource allocation

Risks:

- Storage overcommitment
- Out-of-space errors if many VMs consume full capacity
- Performance degradation

3. Evaluate the security implications of multi-tenancy in cloud datacenters. Analyze how virtualization can both enable and weaken isolation, and recommend mechanisms to strengthen tenant separation.

Answer

Multi-tenancy

Multi-tenancy allows multiple customers (tenants) to share the same physical hardware while keeping their data and applications logically separated.

Benefits of Virtualization

- Strong isolation between virtual machines

- Efficient resource sharing
- Lower infrastructure cost
- Independent operating systems
- Flexible workload management

Security Risks

1. VM Escape
 - Malware escapes from a VM to the hypervisor.
2. Side-channel attacks
 - Attackers obtain sensitive information through shared CPU cache or memory.
3. Data leakage
 - Improper isolation exposes confidential data.
4. Resource exhaustion
 - One tenant consumes excessive resources affecting others.
5. Hypervisor attacks
 - Compromising the hypervisor affects every hosted VM.

Mechanisms to Improve Tenant Separation

- Strong hypervisor security
- Secure VM isolation
- Network segmentation using VLANs
- Virtual firewalls
- Encryption of stored and transmitted data
- Multi-factor authentication
- Role-Based Access Control (RBAC)
- Regular security patches
- Continuous monitoring and intrusion detection

Conclusion

Virtualization improves cloud efficiency but introduces security challenges. Proper isolation, secure hypervisors, encryption, and access control significantly enhance tenant security.

4. Analyze the role of Service-Oriented Architecture (SOA) in enabling interoperability across heterogeneous cloud platforms. Identify key challenges such as service latency, security, and version control, and propose solutions.

Answer

Service-Oriented Architecture (SOA)

SOA is an architectural approach where software functions are provided as reusable services that communicate over standard protocols.

Role in Cloud Computing

- Enables communication between different cloud platforms
- Promotes software reuse
- Supports platform independence
- Simplifies application integration
- Improves scalability and flexibility

Challenges

1. Service Latency

Problem:

- Network delays increase response time.

Solution:

- Load balancing
- Caching
- Content Delivery Networks (CDNs)

2. Security

Problem:

- Unauthorized access and data theft.

Solution:

- HTTPS
- Authentication
- Authorization
- Encryption
- API gateways

3. Version Control

Problem:

- Different versions of services create compatibility issues.

Solution:

- API versioning
- Backward compatibility
- Proper documentation

Conclusion

SOA improves interoperability among heterogeneous cloud systems. Effective security, optimized communication, and proper version management ensure reliable cloud services.

5. A cloud system implementing presence services faces scalability issues during peak usage. Analyze the architectural bottlenecks and recommend design improvements to ensure real-time updates and efficient resource utilization.

Answer

Presence Services

Presence services continuously monitor and update user status (online, offline, busy, available) in real time.

Architectural Bottlenecks

1. Single server overload
2. High network traffic
3. Database bottleneck
4. Large number of simultaneous client connections
5. Delayed status synchronization

Recommended Design Improvements

- Horizontal scaling using multiple servers
- Load balancers to distribute requests
- Distributed databases
- In-memory caching (Redis)

- Publish–Subscribe (Pub/Sub) messaging
- Auto-scaling in cloud environments
- Microservices architecture
- Event-driven communication
- Efficient resource monitoring

Benefits

- Faster real-time updates
- Reduced latency
- High availability
- Better fault tolerance
- Efficient resource utilization
- Improved scalability during peak usage

Criteria	Problem Interpretation (2)	Analytical Reasoning (2.5)	Method Selection (2)	Solution Quality (2)	Presentation of Analysis (1.5)	TOTAL (10)
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

